Naomi Onu

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Education

McMaster University — Sep 2023 - Apr 2028
Bachelor of Science - Honours Mathematics and Statistics Co-op — *Hamilton, Ontario*
Minor in Business, CGPA: 3.7/4.0

Relevant Coursework: Complex Analysis, Data Science, Data Analytics, Mathematical Modelling, Linear Algebra, Operations Management, HR Management, Accounting, Finance

Skills

Coding: Python, R, SQL
Libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-Learn
Developer Tools: GitHub, Kaggle, Jupyter
Analytical Skills: Excel, Power BI, Tableau
Presentation and Design: Canva and PowerPoint
Languages: English and Romanian (Fluent), French (Beginner)

Experience

Data Analyst Intern — Jan. 2026 – Present
Canada Health Infoway — *Toronto, Ontario*

- Led the aggregation, validation, and analysis of national digital health KPIs and performance indicators to support executive and Board-level reporting across provinces and territories.
- Conducted in-depth quantitative analysis of a national Stakeholder Satisfaction Survey, evaluating stakeholder feedback on partnership effectiveness, engagement, collaboration, perceived value, and organizational impact across committees, groups, and digital health intiatives.
- Used Python and Excel to analyze and visualize Canadian Digital Health Survey data, generating insights related to digital health adoption, healthcare access, AI integration, and healthcare utilization trends across Canada.
- Conducted environmental scans and stakeholder research across 10+ Canadian jurisdictions to identify publicly reported digital health indicators and support the development of internal organizational performance frameworks.
- Developed executive-facing dashboards, presentations, and analytical reports to support strategic planning, performance measurement, and data-driven decision-making.

Econometrics Research Assistant — June. 2025 – Mar 2026
McMaster Faculty of Social Sciences — *Hamilton, Ontario*

- Programmed and analyzed a predictive model that determines whether a customer will purchase a product based on multiple factors, using 1,000+ simulated datasets to evaluate purchase probabilities and understand how different covariates influence decision outcomes.
- Processed and transformed large simulated datasets using Python (NumPy, pandas) to build statistical models, extract empirical patterns, and quantify uncertainty in predicted purchase probabilities.
- Applied linear optimization and programming techniques to compute partial effect bounds, integrating model outputs with optimization routines to support identification analysis.
- Designed research-quality matplotlib visualizations to illustrate distributional trends, covariate effects, and sampling variability across thousands of simulated datasets.

Actuarial Pricing Intern — May 2025 – Aug 2025
Northbridge Financial Corporation — *Toronto, Ontario*

- Cleaned, validated, and analyzed large Property and Casualty insurance datasets using Excel and SAS, identifying trends in claims experience and policyholder behavior to support actuarial pricing analyses.
- Assisted in developing pricing models and performing rate-indication analyses to inform premium rate-making decisions across multiple commercial insurance products.
- Collaborated with actuarial and underwriting teams to evaluate key risk factors, interpret loss patterns, and provide data-driven insights for pricing strategy.
- Leveraged WAVE to track, organize, and visualize financial and actuarial results, improving reporting accuracy and supporting senior leadership decision-making.

Projects

BMW Used Car Market Analysis | *Python, Power BI, PowerPoint* | Github May 2026

- Conducted an end-to-end analysis of 10,000+ BMW vehicle listings sourced from Kaggle to identify key drivers of resale pricing, analyze depreciation trends, and develop machine learning models to predict used vehicle prices.
- Cleaned, transformed, and analyzed data in Python using Pandas, NumPy, Matplotlib, and Seaborn, including feature engineering, unit conversions, outlier handling, and exploratory data analysis.
- Developed Linear Regression and Random Forest machine learning models to predict vehicle resale prices, improving predictive performance by 14.6 percentage points in R^2 using Random Forest Regression.
- Built an interactive Power BI dashboard and executive-style PowerPoint highlighting pricing trends, feature importance, depreciation patterns, and predictive modeling insights for business decision-making.